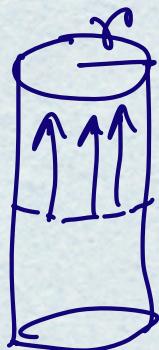


Quiz 3 solution.

Q1.
a)



Surface current density

$$= \frac{I_0}{2\pi r} \boxed{\text{A/m}}$$

In absence of unit deduct 0.5.

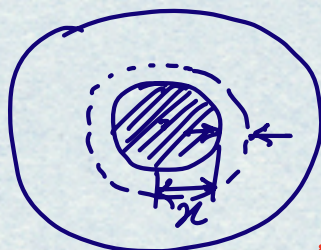
(2)

b)

$$J = \frac{k}{x} \boxed{\text{A/m}^2}$$

[k is a constant]

Deduct 0.5 in absence of unit.



$$dI = J \cdot 2\pi x dx$$

$$= \frac{k}{x} \cdot 2\pi x dx$$

Award 1 mark if someone writes this.

$$\Rightarrow I_0 = \int_0^a k 2\pi dx = 2\pi k a$$

$$\Rightarrow k = \frac{I_0}{2\pi a}$$

$$\therefore J = \frac{I_0}{2\pi a x} \boxed{\text{A/m}^2}$$

deduct 0.5 if unit is absent

(3)

Q2. See tutorial.

- Award 1 mark for finding the EMF/voltage.
- Deduct 1 mark if the final answer doesn't consider rms value.

Q3. (a) $\omega = 1.5\pi \times 10^9$
 $c = 3 \times 10^8 \text{ m/s}$

$$\therefore k = \frac{\omega}{c} = \frac{\frac{3}{2} \times \pi \times 10^9}{3 \times 10^8}$$

$$= \frac{\pi}{2} \times 10^5 = 5\pi$$

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{5\pi} \text{ m} = \frac{2}{5} \text{ m} \quad (\text{or } 0.4 \text{ m})$$

(No part marking)

(b) compare with $e^{i(\omega t - k_x x - k_y y)}$.

$$k_x = 3\pi, k_y = \pi$$

$$k_x^2 + k_y^2 = k^2$$

$$\Rightarrow \pi^2 = 25\pi^2 - 9\pi^2 = 16\pi^2 \Rightarrow \boxed{\pi = 4\pi}$$

1.5 marks till this

$$\begin{aligned}\tan \theta &= \frac{R_y}{R_x} \Rightarrow \theta = \tan^{-1} \left(\frac{R_y}{R_x} \right) \\ &= \tan^{-1} \left(\frac{4}{3} \right) \\ &= \tan^{-1} \left(\frac{4}{3} \right)\end{aligned}$$

Full marks for this.
 [one may simplify $\tan^{-1} \left(\frac{4}{3} \right)$]